

$$\text{---} \blacklozenge \text{---}$$

$$\begin{array}{l} \xi \in \mathbb{R}^d \qquad \qquad \qquad P_\xi = \xi \xi^\top \\ \tilde{e}_t = (1-g_t)h_{t-1} + g_t e_t \qquad \qquad g_t \in (0,1) \\ \|P_\xi \tilde{e}_t\| \leq (1-g_t) \|P_\xi h_{t-1}\| + g_t \|P_\xi e_t\|. \end{array}$$

$$\begin{array}{l} \|P_\xi \tilde{e}_t\| = \|(1-g_t)P_\xi h_{t-1} + g_t P_\xi e_t\| \\ \leq (1-g_t)\|P_\xi h_{t-1}\| + g_t\|P_\xi e_t\|. \end{array}$$

$$\begin{array}{ccc} P_\xi & P_\xi h_{t-1} & P_\xi e_t \end{array}$$

$$g_t \rightarrow 1 \qquad \|P_\xi \tilde{e}_t\| \rightarrow \|P_\xi e_t\| \qquad e_t$$

$$\begin{array}{l} \mathcal{S}_d^+ \\ g_S(u,v) = \mathrm{Tr}(S^{-1}uS^{-1}v), S \in \mathcal{S}_d^+, \; u,v \in T_S\mathcal{S}_d^+. \end{array}$$

$$d_g(S_a,S_b)=\|\log(S_a^{-1/2}S_bS_a^{-1/2})\|_F,$$

$$\log(\cdot)$$

$$\psi: \mathcal{S}_d^+ \rightarrow \mathbb{R}^{d(d+1)/2}, \psi(S) = \mathrm{vech}(\log S),$$

$$\begin{array}{l} \mathrm{vech}(\cdot) \qquad \qquad \qquad \|(\log S_a - \log S_b)\|_F \qquad O(d^3) \\ d{=}128 \quad O(d^3) \approx 2.1{\times}10^6 \qquad < 5\% \\ f_\theta: \mathbb{R}^{d(d+1)/2} \rightarrow \mathbb{R}^{d(d+1)/2} \end{array}$$

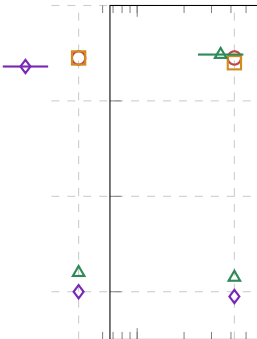
$$f_\theta(x) = W_3 \operatorname{ReLU}(W_2 \operatorname{ReLU}(W_1 x + b_1) + b_2) + b_3,$$

$$\mathcal{L}_{\text{JEPA}} = \mathbb{E}_t \|\hat{s}_{t+1} - \text{sg}(\psi(S_{t+1}))\|_2^2,$$

$$\text{sg}(\cdot) \qquad \qquad \qquad m{=}0.996$$

$$\begin{array}{l} n{=}11 \\ n{=}8 \\ n{=}7 \\ n{=}5 \\ n{=}31 \end{array}$$

$$\chi^2 \qquad \qquad p{=}0.91$$



$$\begin{array}{lll} q_1 & \rightarrow \kappa_1 = 0.8 & g_1 = 0.12 \\ q_2 & \rightarrow \kappa_2 = 1.84 & g_2 = 0.91 \\ q_3 & \rightarrow \kappa_3 = 0.72 & g_3 = 0.08 \\ q_4 & \rightarrow \kappa_4 = 1.97 & g_4 = 0.96 \end{array}$$

$$\begin{array}{ll} \kappa > \kappa_0{=}1.2 & M = 0.028 \\ M = 0.041 & \end{array}$$